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SEMICONDUCTOR DEVICE AND METHOD FOR MANUFACTURING THE SAME

Abstract

A semiconductor device that includes: a base member; a detecting element on the base member and having a first surface with a detector; an insulating protective film covering the detector and the first surface; and a resin package on the base member and including an exposure hole exposing the detector to an outside, a portion of the resin package covering the detecting element such that at least a portion of an outer periphery of the first surface is exposed through the exposure hole, and the resin package includes a recess extending along the portion of the outer periphery exposed through the exposure hole. The detecting element has: a second surface extending from the outer periphery toward the base member, a third surface extending outward from the second surface, and a fourth surface extending from the third surface toward the base member, and wherein the protective film covers the second surface.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International application No. PCT/JP2023/037738, filed Oct. 18, 2023, which claims priority to Japanese Patent Application No. 2022-211859, filed Dec. 28, 2022, the entire contents of each of which are incorporated herein by reference.

TECHNICAL FIELD
[0002] The present disclosure relates to a semiconductor device including a resin package and a method for manufacturing the same.
BACKGROUND ART
[0003] For example, Patent Document 1 discloses a semiconductor device. The semiconductor device includes a detecting element disposed on a base member and having a first surface at which a detector is disposed, and a resin package disposed on the base member and having an exposure hole for allowing the detector to be exposed to the outside. In the semiconductor device, the detecting element has a recess formed at the outer periphery of the first surface.
[0004] In the process of manufacturing the semiconductor device disclosed in Patent Document 1, a wall-shaped portion is formed around the detector by thrusting the detecting element into a release film that closely adheres to the surface of a die. In this structure, the wall-shaped portion reduces the resin material that is to cover the detector during a subsequent operation of filling of the resin material. After the resin material cures to form the resin package, and the die is removed, the portion where the wall-shaped portion is formed serves as the recess. [0005] Patent Document 1: International Publication No. 2021/049138
SUMMARY OF THE DISCLOSURE
[0006] In the semiconductor device disclosed in Patent Document 1, a portion of the detecting element exposed to the outside (a first surface of the detecting element including the detector) is waterproofed by a passivation film (a protective film).
[0007] However, the semiconductor device disclosed in Patent Document 1 has a recess at the outer periphery of an upper surface of the detecting element. This recess may allow a portion of the side surface (the surface extending downward from the outer edge of the first surface) of the detecting element, particularly, a portion located near the first surface of the detecting element to be exposed without being covered with the resin package. Thus, the exposed side surface of the detecting element may come into contact with water. In other words, the detecting element may fail to be waterproofed.
[0008] Thus, the present disclosure aims to solve the above issue, and to provide a semiconductor device capable of reducing degradation of waterproofness of the detecting element, and a method for manufacturing the semiconductor device.

[0009] In order to achieve the above object, the present disclosure has the structure described below.
[0010] A semiconductor device according to an aspect of the present disclosure includes: a base member; a detecting element on the base member and having a first surface with a detector; an insulating protective film that covers the detector and the first surface; and a resin package on the base member and including an exposure hole that exposes the detector of the detecting element to an outside with the protective film interposed therebetween, a portion of the resin package excluding the exposure hole covering the detecting element such that at least a portion of an outer periphery of the first surface is exposed through the exposure hole, and the resin package includes a recess extending along the portion of the outer periphery exposed through the exposure hole, wherein the detecting element further has: a second surface extending from the outer periphery toward the base member, a third surface extending outward from the second surface when viewed in a direction orthogonal to the first surface, and a fourth surface extending from the third surface toward the base member, and wherein the protective film covers the second surface.
[0011] A method for manufacturing a semiconductor device according to an aspect of the present disclosure includes: forming a first groove in a main surface of a base portion to divide the main surface into a plurality of areas; covering the main surface including the first groove with an insulating protective film; cutting the base portion covered with the protective film along the first groove to divide the base portion into a plurality of detecting elements each having a first surface with a detector; placing each of the detecting elements on a base member; adhering a release film to a die having a cavity with a protrusion; placing the die with respect to the base member while the first surface of the detecting elements is thrust into a portion of the release film on a surface of the protrusion; filling the cavity of the die with a resin material; and detaching the die and the release film from a resin package formed from the resin material after curing, wherein at least a portion of an outer periphery of the surface of the protrusion is located outward from an outer periphery of the first surface when viewed in a direction in which the protrusion of the die and the first surface of the detecting element face each other.
[0012] The present disclosure can reduce degradation of waterproofness of the detecting element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0013] FIG. **1** is a perspective view of a semiconductor device according to a first embodiment of the present disclosure.
[0014] FIG. **2** is a top view of the semiconductor device according to the first embodiment of the present disclosure.
[0015] FIG. **3**A is a cross-sectional view of the semiconductor device taken along line A-A in FIG. **2**.
[0016] FIG. **3**B is a cross-sectional view of the semiconductor device taken along line B-B in FIG. **2**.
[0017] FIG. **4** is an enlarged view of a portion enclosed by a broken line in FIG. **3**B.
[0018] FIG. **5**A is a diagram illustrating one process in the manufacture of a detecting element.
[0019] FIG. **5**B is a diagram illustrating one process in the manufacture of the detecting element, following the process in FIG. **5**A.
[0020] FIG. **5**C is a diagram illustrating one process in the manufacture of the detecting element, following the process in FIG. **5**B.
[0021] FIG. **5**D is a diagram illustrating one process in the manufacture of the detecting element, following the process in FIG. **5**C.
[0022] FIG. **6**A is a diagram illustrating one process in the manufacture of a resin package.

[0023] FIG. **6**B is a diagram illustrating one process in the manufacture of the resin package, following the process in FIG. **6**A.
[0024] FIG. **6**C is a diagram illustrating one process in the manufacture of the resin package, following the process in
[0025] FIG. **6**B.
[0026] FIG. **6**D is a diagram illustrating one process in the manufacture of the resin package, following the process in
[0027] FIG. **6**C.
[0028] FIG. **7** is a top view of the semiconductor device, illustrating the positional relationship and the dimensional relationship between a die and the detecting element.
[0029] FIG. **8** is a diagram illustrating the process illustrated in FIG. **6**C, and another cross-sectional view of the semiconductor device.
[0030] FIG. **9** is a top view of a semiconductor device according to a second embodiment of the present disclosure.
[0031] FIG. **10**A is a cross-sectional view of the semiconductor device taken along line A-A illustrated in FIG. **9**.
[0032] FIG. **10**B is a cross-sectional view of the semiconductor device taken along line B-B illustrated in FIG. **9**.
DESCRIPTION OF THE PREFERRED EMBODIMENTS
First Embodiment
[0033] FIG. **1** is a perspective view of a semiconductor device according to a first embodiment of the present disclosure. FIG. **2** is a top view of the semiconductor device according to the first embodiment of the present disclosure. FIG. **3**A and FIG. **3**B are cross-sectional views of the semiconductor device according to the first embodiment, respectively taken along line A-A and line B-B in FIG. **2**. The X-Y-Z orthogonal coordinate system illustrated in these drawings is provided for ease of understanding the present disclosure, not to limit the disclosure.
[0034] As illustrated in FIG. **1** to FIG. **3**, a semiconductor device **10** according to the first embodiment is a pressure sensor that measures the pressure, and includes a base member **12**, a detecting element **14** disposed on the base member **12**, and a protective film **15** that covers a portion of the detecting element **14**. In the first embodiment, a circuit element **16** is disposed on the base member **12**. In the first embodiment, the semiconductor device **10** includes a resin package **18** disposed on the base member **12**. Pressure measured by the semiconductor device **10** includes, for example, absolute pressure, gauge pressure, differential pressure, and atmospheric pressure. The top view in FIG. **2** and top views in FIG. **7** and FIG. **9** described later do not illustrate the protective film **15**.
[0035] As illustrated in FIG. **2**, FIG. **3**A, and FIG. **3**B, the base member **12** is a circuit board having a first main surface **12***a*, and formed from, for example, a ceramic board or a resin board. The base member **12** may be a lead frame. On the first main surface **12***a* of the base member **12**, the detecting element **14** and the circuit element **16** are arranged side by side. In the first embodiment, the detecting element **14** and the circuit element **16** are fixed onto the first main surface **12***a* of the base member **12** with an adhesive material not illustrated. As the adhesive material, a die-attach film or a die-bond material is used.
[0036] As illustrated in FIG. **2** and FIG. **3**A, at the first main surface **12***a* of the base member **12**, connection terminals **12***b* that are electrically connected to the detecting element **14** with bonding wires **20**, and connection terminals **12***c* that are electrically connected to the circuit element **16** with bonding wires **22** are disposed.
[0037] As illustrated in FIG. **3**A and FIG. **3**B, the base member **12** has a second main surface **12***d* opposite to the first main surface **12***a*, and includes an external connection terminal **12***e* on the second main surface **12***d* to be electrically connected to an external electronic device (not illustrated).

[0038] In the first embodiment, the detecting element **14** is a pressure sensor element for measuring pressure. The detecting element **14** is, for example, a piezoresistive pressure sensor element or a capacitive pressure sensor element, and is a micro-electromechanical system (MEMS) device.
[0039] As illustrated in FIG. **3**A, the detecting element **14** has a first main surface **141**, a second main surface **142** opposite to the first main surface **141**, and side surfaces **143**. The side surfaces **143** connect the first main surface **141** and the second main surface **142** to each other. The first main surface **141** is an example of a first surface.
[0040] FIG. **4** is an enlarged view of a portion enclosed by a broken line in FIG. **3**B. As illustrated in FIG. **4**, the side surfaces **143** include a first side surface **143***a*, a second side surface **143***b*, a third side surface **143***c*, a fourth side surface **143***d*, and a fifth side surface **143***e*.
[0041] The first side surface **143***a* extends from an outer periphery **14***f* of the first main surface **141** of the detecting element **14** toward the base member **12**. In the first embodiment, the first side surface **143***a* extends in a Z direction orthogonal to the first main surface **141**, but may instead extend in a direction inclined with respect to the Z direction. The first side surface **143***a* is an example of a second surface. The Z direction is an example of an orthogonal direction.
[0042] The second side surface **143***b* extends outward from the first side surface **143***a* when viewed from above (when viewed in a Z-axis direction), and faces in the Z direction. In the first embodiment, the second side surface **143***b* extends from an end portion of the first side surface **143***a* opposite to the end portion located closer to the outer periphery **14***f* in the Z direction. In the first embodiment, the second side surface **143***b* extends along a virtual plane (a virtual plane orthogonal to the Z direction) extending in an X direction and a Y direction, but may instead extend in a direction inclined with respect to the virtual plane. The second side surface **143***b* is an example of a third surface.
[0043] The third side surface **143***c* extends from the second side surface **143***b* toward the base member **12**. In the first embodiment, the third side surface **143***c* extends from an outer periphery **14***h* of the second side surface **143***b* when viewed from above. In the first embodiment, the third side surface **143***c* extends in the Z direction, but may instead extend in a direction inclined with respect to the Z direction. The third side surface **143***c* is an example of a fourth surface.
[0044] The fourth side surface **143***d* extends outward from the third side surface **143***c* when viewed from above, and faces in the Z direction. In the first embodiment, the fourth side surface **143***d* extends from an end portion of the third side surface **143***c* opposite to the outer periphery **14***h* of the second side surface **143***b* in the Z direction. In the first embodiment, the fourth side surface **143***d* extends along the virtual plane extending in the X direction and the Y direction, but may instead extend in a direction inclined with respect to the virtual plane. The fourth side surface **143***d* is an example of a fifth surface.
[0045] The fifth side surface **143***e* extends from the fourth side surface **143***d* toward the base member **12**. In the first embodiment, the fifth side surface **143***e* extends from an outer periphery **14***i* of the fourth side surface **143***d* when viewed from above. In the first embodiment, the fifth side surface **143***e* extends in the Z direction, but may instead extend in a direction inclined with respect to the Z direction. The fifth side surface **143***e* is an example of a sixth surface.
[0046] Of the side surfaces **143** with the above structure, the second side surface **143***b* serves as a step between the first side surface **143***a* and the third side surface **143***c*. In addition, the fourth side surface **143***d* serves as a step between the third side surface **143***c* and the fifth side surface **143***e*.
[0047] As illustrated in FIG. **2**, the detecting element **14** includes, on the first main surface **141**, multiple connection terminals **14***c* electrically connected to the connection terminals **12***b* of the base member **12** with the bonding wires **20**. Thus, the detecting element **14** is electrically connected to the circuit element **16** with the base member **12** (with a conductor pattern (not illustrated) disposed at the board).
[0048] As illustrated in FIG. **2**, FIG. **3**A, and FIG. **3**B, the detecting element **14** includes a detector **14***d*, on which pressure is exerted, at the first main surface **141**. In the first embodiment, the

detector **14***d* of the detecting element **14** serving as a pressure sensor element is a membrane or a diaphragm that receives pressure.

[0049] As illustrated in FIG. **2** and FIG. **3**A, the detecting element **14** includes a groove **14***e* on the first main surface **141** at a portion between the multiple connection terminals **14***c* and the detector **14***d*, although the reason for this is described later.

[0050] The protective film **15** illustrated in FIG. **3**A, FIG. **3**B, and FIG. **4** is, for example, a passivation film. The protective film **15** contains, for example, silicon nitride (SiN), and has insulating properties. The protective film **15** is provided to reduce, for example, water or dust adhering to the detecting element **14**, or particularly, adhering to the detector **14***d* of the detecting element **14**.

[0051] As illustrated in FIG. **3**A, FIG. **3**B, and FIG. **4**, the protective film **15** covers the detector **14***d*, the first main surface **141**, the first side surface **143***a*, and the second side surface **143***b* of the detecting element **14**. More specifically, the detector **14***d*, the first main surface **141**, the first side surface **143***a*, and the second side surface **143***b* are waterproofed by the protective film **15**.

[0052] In the first embodiment, as illustrated in FIG. **3**B, the circuit element **16** has a first main surface **16***a* and a second main surface **16***b* opposite to the first main surface **16***a*. The circuit element **16** is an element including an application specific integrated circuit (ASIC). In the first embodiment, the circuit element **16** is disposed on the first main surface **12***a* of the base member **12** with the second main surface **16***b* facing the first main surface **12***a*.

[0053] As illustrated in FIG. **2**, the circuit element **16** includes multiple connection terminals **16***c* disposed on the first main surface **16***a*, and electrically connected to the connection terminals **12***c* of the base member **12** with bonding wires **22**. The circuit element **16** is thus electrically connected to the detecting element **14** with the base member **12** (with a conductor pattern (not illustrated) disposed on the board).

[0054] The circuit element **16** is an element that includes a signal processing circuit. The signal processing circuit processes signals output from the detecting element **14**, and outputs the processed signals to the base member **12**. For example, in the first embodiment, the circuit element **16** includes a converter that converts voltage signals output from the detecting element **14** to digital signals, a filter that filters the digital signals from the converter, a temperature sensor that detects the temperature, a processor that corrects the digital signals filtered based on the temperature detected by the temperature sensor, and a memory that stores a correction coefficient used to correct the digital signals using the detected temperature.

[0055] In the first embodiment, the resin package **18** illustrated in FIG. **1** to FIG. **4** is a package formed by shaping a hard resin such as a thermosetting resin on the first main surface **12***a* of the base member **12**. The method for forming the resin package **18** is described in detail later.

[0056] A portion of the first main surface **12***a* of the base member **12** including the multiple connection terminals **12***b* and **12***c* is covered with the resin package **18** to be protected and waterproofed. The detecting element **14** (particularly, the connection terminals **14***c*), the circuit element **16** (particularly, the connection terminal **16***c*), and the bonding wires **20** and **22** that electrically connect these elements to each other are embedded in the resin package **18** to be protected and waterproofed. More specifically, the resin package **18** protects electric connection between the base member **12** and the detecting element **14** and electric connection between the base member **12** and the circuit element **16**.

[0057] As illustrated in FIG. **1**, FIG. **3**A, and FIG. **3**B, to allow pressure to be exerted on the detector **14***d* of the detecting element **14** with the protective film **15** interposed therebetween, the resin package **18** includes an exposure hole **18***a* to allow the detector **14***d* of the detecting element **14** to be exposed to the outside of the resin package **18** with the protective film **15** interposed therebetween.

[0058] More specifically, in the first embodiment, as illustrated in FIG. **1**, the resin package **18** includes a rectangular parallelepiped body portion **18***b* disposed on the base member **12**, and a

cylindrical ring-shaped holding portion **18***c* disposed on the body portion **18***b*. The exposure hole **18***a* is open in a top surface **18***d* of the ring-shaped holding portion **18***c*, and extends toward the first main surface **12***a* of the base member **12**. Pressure is exerted on, with the protective film **15** interposed therebetween, the detector **14***d* of the detecting element **14** exposed through the exposure hole **18***a* to the outside of the resin package **18** with the protective film **15** interposed therebetween, and the semiconductor device **10** can measure the pressure.
[0059] The ring-shaped holding portion **18***c* of the resin package **18** is used to hold an O-ring (not illustrated) on its outer peripheral surface. For example, the ring-shaped holding portion **18***c* of the resin package **18** of the semiconductor device **10** is inserted, with the O-ring interposed therebetween, into a through-hole that connects the outside and the internal space in the housing of an electronic device to which the semiconductor device **10** is attached. More specifically, in the first embodiment, the resin package **18** also has a function of attaching the semiconductor device **10** to an electronic device.
[0060] As illustrated in FIG. **2**, to allow at least a portion of the outer periphery **14***f* of the first main surface **141** of the detecting element **14** to be exposed through the exposure hole **18***a* of the resin package **18**, the resin package **18** includes a recess **18***e* extending along the exposed portion of the outer periphery **14***f*. More specifically, the portion of the resin package **18** disposed on the first main surface **141** of the detecting element **14** is kept to a minimum. The reason why the resin package **18** is disposed in this manner is described later.
[0061] More specifically, in the first embodiment, the outer periphery **14***f* of the first main surface **141** of the detecting element **14** has a shape of a rectangle with four sides when viewed from above. One side **14***g* of the outer periphery **14***f* along which the multiple connection terminals **14***c* connected to the bonding wires **20** are arranged adjacent to each other is embedded in the resin package **18** together with the connection terminals **14***c* and the bonding wires **20**. Other sides of the outer periphery **14***f* other than the side **14***g* are exposed to the outside of the resin package **18**. The resin package **18** includes the recess **18***e* extending along the sides of the outer periphery **14***f* other than the side **14***g*.
[0062] The recess **18***e* has a bottom portion shaped to be located closer to the base member **12** than the first main surface **141** of the detecting element **14**.
[0063] As illustrated in FIG. **4**, a depth D of the recess **18***e* in the Z direction is longer than a length L of the first side surface **143***a* in the Z direction. In the first embodiment, the depth D of the recess **18***e* is, for example, approximately 10 μm. In the first embodiment, the length L of the first side surface **143***a* is larger than or equal to 1 μm and smaller than or equal to 10 μm. The depth D of the recess **18***e* may be smaller than or equal to the length L of the first side surface **143***a*. Alternatively, the depth D of the recess **18***e* and the length L of the first side surface **143***a* are not limited to the above numerical values and the above numerical ranges.
[0064] The reason why the length L of the first side surface **143***a* is set to be larger than or equal to 1 μm is because the portion of the first side surface **143***a* is left without being covered with the resin package **18** by approximately 1 μm during the manufacturing process of the semiconductor device **10**. Here, the portion of the first side surface **143***a* left without being covered with the resin package **18** is a portion of the first side surface **143***a* located closer to the first main surface **141** in the Z direction than a boundary **14***j* in FIG. **4**.
[0065] The reason why the length L of the first side surface **143***a* is set to be smaller than or equal to 10 μm is as follows. If the length L is larger than 10 μm, the protective film **15** is more likely to fail to be formed over the entirety of the first side surface **143***a* during the manufacturing process of the detecting element **14**, and the third side surface **143***c* and the fourth side surface **143***d* are more likely to fail to be formed by forming second grooves **64** in a second groove forming process described later.
[0066] As illustrated in FIG. **4**, when viewed in a direction along the first main surface **141**, in other words, when viewed in a direction along the XY plane, an inclination angle θ**1** is larger than

an inclination angle θ**2**. The inclination angle θ**1** is an angle of a virtual tangent **41** with respect to the first side surface **143***a*. The virtual tangent **41** is a tangent to the surface of the recess **18***e* at the boundary **14***j* between the recess **18***e* and the protective film **15** covering the first side surface **143***a*. The inclination angle θ**2** is an angle of a virtual line **42** with respect to the first side surface **143***a*. The virtual line **42** is a line passing the outer periphery **14***f*, which is a boundary between the first main surface **141** and the first side surface **143***a*, and the outer periphery **14***h*, which is a boundary between the second side surface **143***b* and the third side surface **143***c*.
[0067] When viewed in a direction along the XY plane, the inclination angle θ**1** is larger than an inclination angle θ**3**. The inclination angle θ**3** is an angle of a virtual line **43** with respect to the first side surface **143***a*. The virtual line **43** is a line passing the outer periphery **14***f*, which is a boundary between the first main surface **141** and the first side surface **143***a*, and the outer periphery **14***i*, which is a boundary between the fourth side surface **143***d* and the fifth side surface **143***e*.
[0068] In the first embodiment, the inclination angle θ**1** is larger than or equal to 35 degrees, but may instead be smaller than 35 degrees.
[0069] In the first embodiment, the direction along the first main surface **141** (the direction along the XY plane) is any direction along the first main surface **141**. More specifically, the relationship between the inclination angles θ**1**, θ**2**, and θ**3** described above is satisfied regardless of when viewed in any direction along the first main surface **141**. The relationship between the inclination angles θ**1**, θ**2**, and θ**3** described above may be satisfied only when viewed in one or more of the directions along the first main surface **141**. Alternatively, the relationship between the inclination angles θ**1**, θ**2**, and θ**3** described above may fail to be satisfied regardless of any direction along the first main surface **141**.
[0070] In the resin package **18** with the above structure, the multiple connection terminals **14***c* and portions around the connection terminals **14***c* at the first main surface **141** of the detecting element **14** are embedded in the resin package **18**.
[0071] When the resin package **18** is adjacent to the outer periphery **14***f* of the first main surface **141** (more specifically, the protective film **15** covering the outer periphery **14***f*), the resin package **18** completely covers the first side surface **143***a*.
[0072] However, when the resin package **18** is adjacent to the first side surface **143***a* (more specifically, the protective film **15** covering the first side surface **143***a*) at a position closer to the base member **12** than the outer periphery **14***f* of the first main surface **141**, the resin package **18** fails to cover at least a portion of the first side surface **143***a*. More specifically, in the semiconductor device **10**, the resin package **18** does not cover at least a portion of the first side surface **143***a*. In this structure, the first side surface **143***a* can be exposed through the exposure hole **18***a*. However, the first side surface **143***a* is covered with the protective film **15**, and the first side surface **143***a* is thus protected from adhesion of, for example, water or dust. In the first embodiment, the resin package **18** does not cover a portion of the first side surface **143***a* (more specifically, a portion of the first side surface **143***a* located closer to the first main surface **141** in the Z direction from the boundary **14***j*).
[0073] When the resin package **18** is adjacent to the second side surface **143***b* (more specifically, the protective film **15** covering the second side surface **143***b*), the resin package **18** does not cover the entirety of the first side surface **143***a* and at least a portion of the second side surface **143***b*. More specifically, the semiconductor device **10** has a structure where the resin package **18** does not cover the entirety of the first side surface **143***a* and at least a portion of the second side surface **143***b*. In this case, the first side surface **143***a* and the second side surface **143***b* can be exposed through the exposure hole **18***a*. However, the first side surface **143***a* and the second side surface **143***b* are covered with the protective film **15**, and the first side surface **143***a* is thus protected from adhesion of, for example, water or dust. In the first embodiment, the resin package **18** covers the entirety of the second side surface **143***b*.
[0074] As described above, the resin package **18** covers the detecting element **14** with a portion

excluding the exposure hole **18***a*. The portion excluding the exposure hole **18***a* corresponds to the first main surface **141** and the detector **14***d*. However, depending on the structure of the resin package **18**, the portion excluding the exposure hole **18***a* can include, in addition to the first main surface **141** and the detector **14***d*, at least one of the first side surface **143***a* and the second side surface **143***b*.

[0075] Subsequently, a method for manufacturing the semiconductor device **10** with the above structure, particularly, a method for manufacturing the detecting element **14** and the resin package **18** is described.

[0076] FIG. **5**A to FIG. **5**D each illustrate one process of manufacturing a detecting element in the method for manufacturing the semiconductor device **10**. In FIG. **5**A to FIG. **5**D, the detector **14***d* is not illustrated.

[0077] First, a base portion **60** (refer to FIG. **5**A to FIG. **5**D) is prepared. The base portion **60** serves as a base of the detecting element **14** formed from a MEMS device.

[0078] As illustrated in FIG. **5**A, a first groove forming process for forming first grooves **62** in a main surface **61** of the base portion **60** is performed. Thus, the main surface **61** is divided into multiple areas **63** with the first grooves **62** in a plan view of the base portion **60** in which the main surface **61** is viewed in a direction orthogonal to the main surface **61**, more specifically, when viewed from above (when viewed in the Z-axis direction). In the first embodiment, the main surface **61** is divided into the multiple areas **63** arrayed in the X direction and the Y direction, and each area **63** is rectangular. In the first groove forming process, the groove **14***e* (refer to FIG. **2** and FIG. **3**A) may be formed in each area **63** of the main surface **61** of the base portion **60** between the first groove forming process and a protective film forming process described below.

[0079] Subsequently, as illustrated in FIG. **5**B, a protective film forming process for forming the insulating protective film **15** on the main surface **61** is performed. The main surface **61** includes the first grooves **62** formed in the first groove forming process. Thus, the main surface **61** including the first grooves **62** is covered with the protective film **15**.

[0080] Subsequently, as illustrated in FIG. **5**C, a second groove forming process for forming the second grooves **64** along bottom surfaces **62***a* of the first grooves **62** is performed. A width W**1** of a bottom surface **64***a* of each second groove **64** is smaller than a width W**2** of the bottom surface **62***a* of each first groove **62**. Thus, a step is formed between each first groove **62** and the corresponding second groove **64** by the difference between the two widths W**1** and W**2**.

[0081] Subsequently, as illustrated in FIG. **5**D, a cutting process for cutting the base portion **60** covered with the protective film **15** along the first grooves **62** and the second grooves **64** is performed. In the first embodiment, the base portion **60** is cut in the Z direction from the bottom surfaces **64***a* of the second grooves **64**. Thus, the base portion **60** is divided into multiple detecting elements **14**. In the first embodiment, a blade (not illustrated) for cutting the base portion **60** has a width smaller than the width W**2** of the bottom surfaces **64***a* of the second grooves **64**.

[0082] In each of the multiple detecting elements **14** divided in the cutting process, the portion corresponding to a side surface **62***b* of each first groove **62** serves as the first side surface **143***a*. The portion corresponding to the bottom surface **64***a* of each first groove **62** serves as the second side surface **143***b*. The portion corresponding to the side surface **64***b* of each second groove **64** serves as the third side surface **143***c*. The portion corresponding to the bottom surface **64***a* of each second groove **64** serves as the fourth side surface **143***d*. A cut surface exposed after the base portion **60** is cut in the cutting process serves as the fifth side surface **143***e*.

[0083] The manufacture of the base portion **60** and the above processes are performed with a known fine-processing technology (for example, a technology for manufacturing a semiconductor integrated circuit). For example, when the base portion **60** includes multiple layers, the first grooves **62** and the second grooves **64** are formed by, for example, etching during a process of laminating layers of the base portion **60**. In this case, in the process of laminating layers of the base portion **60**, the first grooves **62** may be formed after the second grooves **64** are formed.

[0084] FIG. **6**A to FIG. **6**D each illustrate a process of manufacturing a resin package in a method for manufacturing the semiconductor device **10**.
[0085] As illustrated in FIG. **6**A, the detecting element **14** formed in the above manner and the circuit element **16** are disposed on the base member **12**. More specifically, the detecting element **14** and the circuit element **16** are mounted on the base member **12**.
[0086] The base member **12** at which the bonding wires **20** and **22** are disposed is then placed in a predetermined position with respect to a die **50**. The die **50** includes a cavity **50***a* for forming the resin package **18** and suction holes **50***b* for sucking air in the cavity **50***a*. The cavity **50***a* also includes a protrusion **50***c* for forming the exposure hole **18***a* of the resin package **18**. The protrusion **50***c* has a flat top surface **50***d*.
[0087] Before the die **50** is filled with the resin material of the resin package **18**, a release film **52** is sucked through the suction holes **50***b* to come into close contact with the surface of the cavity **50***a*. The release film **52** is, for example, a heat-resistant resin film having a surface coated with a release agent.
[0088] As illustrated in FIG. **6**B, the die **50** in which the release film **52** is in close contact with the surface of the cavity **50***a* moves toward the base member **12**, and the top surface **50***d* of the protrusion **50***c* of the die **50** comes into contact with the detecting element **14** on the base member **12** with the release film **52** interposed therebetween. Thus, the die **50** is placed with respect to the base member **12** while the first main surface **141** of the detecting element **14** is thrust into a portion of the release film **52** on the top surface **50***d* of the protrusion **50***c*.
[0089] FIG. **7** is a top view of a semiconductor device illustrating the positional relationship between the die and the detecting element.
[0090] As illustrated in FIG. **7**, in the first embodiment, the top surface **50***d* of the protrusion **50***c* of the die **50** is rectangular, and has an outer periphery **50***e*. As illustrated in FIG. **6**B, when the top surface **50***d* of the protrusion **50***c* of the die **50** is in contact with the detecting element **14** with the release film **52** interposed therebetween, at least a portion of the outer periphery **50***e* of the top surface **50***d* is located outward from the outer periphery **14***f* of the first main surface **141** when viewed in a direction in which the protrusion **50***c* and the first main surface **141** of the detecting element **14** face each other (when viewed in the Z-axis direction).
[0091] In the first embodiment, as illustrated in FIG. **7**, a portion of the outer periphery **50***e* of the top surface **50***d* of the protrusion **50***c* of the die **50** is located outward from a portion of the outer periphery **14***f* of the first main surface **141** of the detecting element **14** excluding the side **14***g* along which the multiple connection terminals **14***c* are arranged. More specifically, a portion near the side **14***g* and the connection terminals **14***c* does not face the top surface **50***d* of the protrusion **50***c* of the die **50**.
[0092] With the positional and dimensional relationships between the top surface **50***d* of the protrusion **50***c* of the die **50** and the first main surface **141** of the detecting element **14**, as illustrated in FIG. **6**B, a wall-shaped portion **52***a* of the release film **52** that protrudes toward the base member **12** beyond the first main surface **141** and surrounds a portion of the outer periphery **14***f* of the first main surface **141** is formed.
[0093] As illustrated in FIG. **6**C, the cavity **50***a* of the die **50** is filled with a molten resin material **54**. At this time, the wall-shaped portion **52***a* of the release film **52** blocks the resin material **54** that is to enter a space between the release film **52** and the first main surface **141** of the detecting element **14**. This structure thus reduces the resin material **54** that covers the detector **14***d* of the detecting element **14**.
[0094] However, in the first embodiment, as illustrated in FIG. **8**, the wall-shaped portion **52***a* of the release film **52** is not located at the side **14***g* of the outer periphery **14***f* of the first main surface **141** of the detecting element **14**. Thus, the resin material **54** may enter a space between the release film **52** and the first main surface **141** of the detecting element **14** around the side **14***g*.
[0095] To address this, the first embodiment includes the groove **14***e* disposed near a portion

between the detector **14***d* and the multiple connection terminals **14***c* on the first main surface **141** of the detecting element **14**. When the resin material **54** is to move from the side **14***g* toward a space between the release film **52** and the first main surface **141** of the detecting element **14**, the moving resin material **54** flows into the groove **14***e* before reaching the detector **14***d*. This structure thus reduces the moving resin material **54** that reaches the detector **14***d* of the detecting element **14** and that covers at least a portion of the detector **14***d*.

[0096] When the multiple connection terminals **14***c* and the detector **14***d* are spaced a sufficiently large distance from one another, more specifically, when the multiple connection terminals **14***c* and the detector **14***d* are spaced a distance that does not allow the moving resin material **54** to reach the detector **14***d*, the groove **14***e* may be eliminated.

[0097] As illustrated in FIG. **6**C, after the cavity **50***a* of the die **50** is filled with the resin material **54** and then the resin material **54** cures, as illustrated in FIG. **6**D, the die **50** and the release film **52** are detached from the cured resin material **54**. The resin package **18** including the recess **18***e* having a shape corresponding to the wall-shaped portion **52***a* of the release film **52** illustrated in FIG. **6**C and following a part of the outer periphery **14***f* of the first main surface **141** of the detecting element **14** is thus formed on the base member **12**.

[0098] According to the first embodiment described above, the semiconductor device **10** including the detecting element **14** including the detector **14***d* and the resin package **18** including the exposure hole **18***a* that allows the detector **14***d* to be exposed to the outside can reduce degradation of detection performance of the detecting element **14** due to the resin package **18**.

[0099] More specifically, as illustrated in FIG. **2**, to allow at least a portion of the outer periphery **14***f* of the first main surface **141** of the detecting element **14** to be exposed through the exposure hole **18***a*, the resin package **18** includes the recess **18***e* extending along the exposed outer periphery **14***f*. Thus, first, the resin material **54** of the resin package **18** is prevented from covering the detector **14***d* of the detecting element **14** during manufacture of the semiconductor device **10**. This structure thus reduces degradation of detection performance of the detecting element **14** due to the resin package **18**.

[0100] Thus, a portion of the resin package **18** disposed on the first main surface **141** of the detecting element **14** is kept to a minimum. This structure thus reduces degradation of detection performance of the detecting element **14**.

[0101] More specifically, during use of the semiconductor device **10**, the resin package **18** may receive an external force. For example, in the first embodiment, the resin package **18** receives compressive force from an O-ring held by the ring-shaped holding portion **18***c* of the resin package **18**. The external force acts on the detecting element **14** with the resin package **18** interposed therebetween. Particularly, when a portion of the resin package **18** is disposed on the first main surface **141** of the detecting element **14**, the first main surface **141** may receive distortion with the resin package **18** interposed therebetween. Thus, the detector **14***d* disposed on the first main surface **141** may be deformed, and the detection performance of the detecting element **14** may be degraded.

[0102] Thus, to allow at least a portion of the outer periphery **14***f* of the first main surface **141** of the detecting element **14** to be exposed through the exposure hole **18***a*, the resin package **18** includes the recess **18***e* extending along the exposed outer periphery **14***f*. This structure reduces occurrence of distortion in the first main surface **141** of the detecting element **14** (compared to the structure where the entirety of the outer periphery **14***f* is embedded in the resin package **18** and the recess **18***e* is thus not formed in the resin package **18**). This structure thus reduces degradation of the detection performance of the detecting element **14** due to the resin package **18**.

[0103] In the first embodiment, the recess **18***e* is formed along the outer periphery **14***f*. Thus, compared to the second side surface **143***b* and the third side surface **143***c*, the first side surface **143***a* located near the outer periphery **14***f* is more likely to be exposed by the recess **18***e* without being covered with the resin package **18**. In the first embodiment, the protective film **15** covers the first side surface **143***a* in addition to the first main surface **141**. More specifically, in the first embodiment, the first side surface **143***a* that is highly likely to be exposed is covered with the protective film **15**. This structure can thus reduce the likelihood of the first side surface **143***a* coming into contact with water. This structure can thus reduce degradation of waterproofness of the detecting element **14**.

[0104] In the first embodiment, the protective film **15** covers the second side surface **143***b* in addition to the first main surface **141** and the first side surface **143***a*. This structure can thus reduce the likelihood of the second side surface **143***b* coming into contact with water. This structure can thus reduce degradation of waterproofness of the detecting element **14** compared to a structure in which the second side surface **143***b* is not covered with the protective film **15**.

[0105] When the length L of the first side surface **143***a* in the Z direction is excessively long, the protective film **15** may fail to be formed over the entirety of the first side surface **143***a* in the manufacturing process of the semiconductor device **10**. In the first embodiment, the length L of the first side surface **143***a* in the Z direction is shorter than the depth D of the recess **18***e* in the Z direction. Thus, compared to a structure where the length L is larger than or equal to the depth D, in this structure, the protective film **15** is less likely to fail to be formed over the entirety of the first side surface **143***a*.

[0106] In the first embodiment, the inclination angle θ**1** is larger than the inclination angle θ**2**. Thus, the surface defining the recess **18***e* of the resin package **18**, which covers the first side surface **143***a*, the second side surface **143***b*, and the third side surface **143***c*, that is, the surface of the resin package **18** is spaced farther apart from the first side surface **143***a*, the second side surface **143***b*, and the third side surface **143***c* as it extends toward the base member **12**. Thus, the first side surface **143***a*, the second side surface **143***b*, and the third side surface **143***c* are less likely to be exposed through the surface defining the recess **18***e* of the resin package **18** (the surface of the resin package **18**).

[0107] In the first embodiment, to manufacture, by cutting the base portion **60** on which the protective film **15** is disposed, the multiple detecting elements **14** on each of which the protective film **15** is disposed, the cutting is started at the fourth side surface **143***d* to form the fifth side surface **143***e* as a cut surface. Thus, regardless of when the detecting element **14** is chipped at the fourth side surface **143***d* or the fifth side surface **143***e*, the first side surface **143***a* is less likely to be affected by the chipping. Thus, during the cutting, the protective film **15** disposed on the first side surface **143***a* is less likely to be broken.

[0108] In the first embodiment, the inclination angle θ**1** is larger than the inclination angle θ**3**. Thus, the surface defining the recess **18***e* of the resin package **18**, which covers the fourth side surface **143***d* and the fifth side surface **143***e*, that is, the surface of the resin package **18** is spaced farther apart from the fourth side surface **143***d* and the fifth side surface **143***e* as it extends toward the base member **12**. Thus, the fourth side surface **143***d* and the fifth side surface **143***e* are less likely to be exposed through the surface defining the recess **18***e* of the resin package **18** (the surface of the resin package **18**).

[0109] With this manufacturing method, when the base portion **60** is cut and divided into the multiple detecting elements **14**, the side surface of each first groove **62** covered with the protective film **15** serves as the first side surface **143***a*, and the bottom surface of each first groove **62** covered with the protective film **15** serves as the second side surface **143***b*. In this manner, the semiconductor device **10** including the detecting element **14** in which, in addition to the first main surface **141**, the first side surface **143***a* and the second side surface **143***b* are covered with the protective film **15** can be easily manufactured.

[0110] With this manufacturing method, regardless of when the base portion **60** is chipped while the base portion **60** is cut along the second grooves **64**, the first grooves **62** different from the second grooves **64** are less likely to be affected by the chipping. Thus, the protective film covering the first grooves **62** is less likely to be broken by chipping.

Second Embodiment

[0111] As illustrated in FIG. **2**, in the first embodiment, a portion (the side **14***g*) of the outer periphery **14***f* of the first main surface **141** of the detecting element **14** is embedded in the resin package **18**. Thus, the recess **18***e* of the resin package **18** is not located throughout the entire outer periphery **14***f* of the first main surface **141** of the detecting element **14**. Unlike this structure, in the second embodiment, the recess of the resin package is located throughout the entire outer periphery of the first surface of the detecting element. A semiconductor device according to the second embodiment is described mainly in terms of this different point.
[0112] FIG. **9** is a top view of a semiconductor device according to the second embodiment. FIG. **10**A and FIG. **10**B are cross-sectional views of the semiconductor device according to the second embodiment taken along line C-C and line D-D in FIG. **9**.
[0113] As illustrated in FIG. **9**, FIG. **10**A, and FIG. **10**B, in a semiconductor device **110** according to the second embodiment, a detecting element **114** includes multiple connection terminals **114***c* disposed not at a first surface **1141** on which a detector **114***d* is disposed, but at a second surface **1142** opposite to the first surface **1141**. The detecting element **114** has side surfaces **1143**. The side surfaces **1143** correspond to the side surfaces **143** of the detecting element **14** according to the first embodiment.
[0114] The detecting element **114** is placed on a base member **112** with the second surface **1142** facing the base member **112**. The multiple connection terminals **114***c* on the second surface **1142** are electrically connected to the base member **112** (the conductor pattern on a first surface **112***a*) with solder **120**.
[0115] In the detecting element **114** with this structure, no connection terminal is disposed at the first surface **1141**, and a resin package **118** has no need of covering the first surface **1141** of the detecting element **114**. Thus, as illustrated in FIG. **9**, to expose the entirety of an outer periphery **114***f* of the first surface **1141** of the detecting element **114** through an exposure hole **118***a*, the resin package **118** includes an annular recess **118***e* extending along the exposed outer periphery **114***f*.
[0116] In the second embodiment, a circuit element **116** also includes multiple connection terminals **116***c* disposed at a second surface **116***b* facing the base member **112**. The multiple connection terminals **116***c* are electrically connected to the base member **112** with solder **122**.
[0117] As in the case of the first embodiment described above, the semiconductor device **110** according to the second embodiment including the detecting element **114** including the detector **114***d* and the resin package **118** having the exposure hole **118***a* through which the detector **114***d* is exposed to the outside can also reduce degradation of detection performance of the detecting element **114** due to the resin package **118**.
[0118] Although the present disclosure has been described using multiple embodiments, embodiments of the present disclosure are not limited to these.
[0119] For example, in the first embodiment described above, as illustrated in FIG. **2** and FIG. **3**B, the detecting element **14** and the circuit element **16** are mounted on the base member **12** while being arranged side by side. However, embodiments of the present disclosure are not limited to this example.
[0120] For example, the circuit element may be mounted on the base member, and the detecting element may be mounted on the circuit element. More specifically, the detecting element is indirectly disposed on the base member with the circuit element interposed therebetween. In this case, for example, the detecting element may be electrically connected to the circuit element with bonding wires or solder.
[0121] Alternatively, the detecting element and the circuit element may be integrated. For example, a circuit in the circuit element may be incorporated in the detecting element.
[0122] In the first embodiment described above, the semiconductor device **10** is a so-called pressure sensor that measures pressure. However, embodiments of the present disclosure are not limited to this example. For example, a semiconductor device according to an embodiment of the present disclosure may be a sensor that detects (measures), for example, light, ultrasonic waves, or

a specific gas, or may be a microphone. More specifically, a semiconductor device according to an embodiment of the present disclosure includes a detecting element that includes a detector, which is exposed to the outside of the semiconductor device to be capable of detecting a detection target. For example, when the semiconductor device is an optical sensor that detects light, the optical sensor includes a photodiode as the detecting element. Alternatively, for example, when the semiconductor device is an ultrasonic wave sensor that detects ultrasonic waves, the ultrasonic wave sensor includes an ultrasonic transducer as the detecting element.

[0123] The semiconductor device described above can be expressed in the following manner.

[0124] A semiconductor device according to a first aspect, comprising: a base member; a detecting element on the base member and having a first surface with a detector; an insulating protective film that covers the detector and the first surface; and a resin package on the base member and including an exposure hole that exposes the detector of the detecting element to an outside with the protective film interposed therebetween, a portion of the resin package excluding the exposure hole covering the detecting element such that at least a portion of an outer periphery of the first surface is exposed through the exposure hole, and the resin package includes a recess extending along the portion of the outer periphery exposed through the exposure hole, wherein the detecting element further has: a second surface extending from the outer periphery toward the base member, a third surface extending outward from the second surface when viewed in a direction orthogonal to the first surface, and a fourth surface extending from the third surface toward the base member, and wherein the protective film covers the second surface.

[0125] A semiconductor device according to a second aspect is the semiconductor device according to the first aspect, wherein the protective film further covers the third surface.

[0126] A semiconductor device according to a third aspect is the semiconductor device according to the first or second aspect, wherein a length of the second surface in the direction orthogonal to the first surface is shorter than a depth of the recess in the direction orthogonal to the first surface.

[0127] A semiconductor device according to a fourth aspect is the semiconductor device according to any one of the first to third aspects, wherein, when viewed in a direction along the first surface, a first inclination angle of a virtual tangent to a surface of the recess at a first boundary between the recess and the protective film that covers the second surface with respect to the second surface is larger than a second inclination angle of a first virtual line passing a second boundary between the first surface and the second surface and a third boundary between the third surface and the fourth surface with respect to the second surface.

[0128] A semiconductor device according to a fifth aspect is the semiconductor device according to any one of the first to fourth aspects, wherein the detecting element further has a fifth surface extending outward from the fourth surface when viewed in the direction orthogonal to the first surface, and a sixth surface extending from the fifth surface toward the base member.

[0129] A semiconductor device according to a sixth aspect is the semiconductor device according to the fifth aspect, wherein, when viewed in a direction along the first surface, the first inclination angle of the virtual tangent to the surface of the recess at the first boundary between the recess and the protective film that covers the second surface with respect to the second surface is larger than a third inclination angle of a second virtual line passing the second boundary between the first surface and the second surface and a fourth boundary between the fifth surface and the sixth surface with respect to the second surface.

[0130] A method for manufacturing a semiconductor device according to a seventh aspect, the method comprising: forming a first groove in a main surface of a base portion to divide the main surface into a plurality of areas; covering the main surface including the first groove with an insulating protective film; cutting the base portion covered with the protective film along the first groove to divide the base portion into a plurality of detecting elements each having a first surface with a detector; placing each of the detecting elements on a base member; adhering a release film to a die having a cavity with a protrusion; placing the die with respect to the base member while the first surface of the detecting elements is thrust into a portion of the release film on a surface of the protrusion; filling the cavity of the die with a resin material; and detaching the die and the release film from a resin package formed from the resin material after curing, wherein at least a portion of an outer periphery of the surface of the protrusion is located outward from an outer periphery of the first surface when viewed in a direction in which the protrusion of the die and the first surface of the detecting element face each other.

[0131] A method for manufacturing a semiconductor device according to an eight aspect is the method for manufacturing a semiconductor device according to the seventh aspect, further comprising: after covering the main surface with the protective film and before cutting the base portion, forming a second groove with a smaller width than a bottom surface of the first groove along the bottom surface; and cutting the base portion covered with the protective film along the first groove and the second groove so as to divide the base portion into the plurality of detecting elements.

[0132] By combining any two or more of the various embodiments and modifications described above as appropriate, effects of each of the embodiments and modifications can be exerted.

[0133] The present disclosure is fully described in relation to preferred embodiments with reference to the drawings as appropriate, but various modifications and alterations are apparent to persons having ordinary skill in the art. Such modifications and alterations are to be understood to be included within the scope of the present disclosure as long as they do not depart from the scope defined by the appended claims.

INDUSTRIAL APPLICABILITY

[0134] The present disclosure is applicable to a semiconductor device including a detecting element including a detector and a resin package that allows the detector to be exposed to the outside.

REFERENCE SIGNS LIST

[0135] **10** semiconductor device [0136] **12** base member [0137] **14** detecting element [0138] **141** first main surface (first surface) [0139] **143***a* first side surface (second surface) [0140] **143***b* second side surface (third surface) [0141] **143***c* third side surface (fourth surface) [0142] **143***d* fourth side surface (fifth surface) [0143] **143***e* fifth side surface (sixth surface) [0144] **14***d* detector [0145] **14***f* outer periphery (boundary) [0146] **14***h* outer periphery (boundary) [0147] **14***i* outer periphery (boundary) [0148] **14***j* boundary [0149] **15** protective film [0150] **18** resin package [0151] **18***a* exposure hole [0152] **18***e* recess [0153] **41** virtual tangent [0154] **42** virtual line [0155] **43** virtual line [0156] **50** die [0157] **50***a* cavity [0158] **50***c* protrusion [0159] **50***d* top surface [0160] **52** release film [0161] **54** resin material [0162] **60** base portion [0163] **61** main surface [0164] **62** first groove [0165] **62***a* bottom surface [0166] **63** area [0167] **64** second groove [0168] D depth [0169] L length [0170] θ**1** inclination angle [0171] θ**2** inclination angle [0172] θ**3** inclination angle

Claims

1. A semiconductor device, comprising: a base member; a detecting element on the base member and having a first surface with a detector; an insulating protective film that covers the detector and the first surface; and a resin package on the base member and including an exposure hole that exposes the detector of the detecting element to an outside with the protective film interposed therebetween, a portion of the resin package excluding the exposure hole covering the detecting element such that at least a portion of an outer periphery of the first surface is exposed through the exposure hole, and the resin package includes a recess extending along the portion of the outer periphery exposed through the exposure hole, wherein the detecting element further has: a second surface extending from the outer periphery toward the base member, a third surface extending outward from the second surface when viewed in a direction orthogonal to the first surface, and a fourth surface extending from the third surface toward the base member, and wherein the protective film covers the second surface.

2. The semiconductor device according to claim 1, wherein the protective film further covers the third surface.

3. The semiconductor device according to claim 1, wherein a length of the second surface in the direction orthogonal to the first surface is shorter than a depth of the recess in the direction orthogonal to the first surface.

4. The semiconductor device according to claim 3, wherein the length of the second surface in the direction orthogonal to the first surface is 1 μm to 10 μm.

5. The semiconductor device according to claim 1, wherein, when viewed in a direction along the first surface, a first inclination angle of a virtual tangent to a surface of the recess at a first boundary between the recess and the protective film that covers the second surface with respect to the second surface is larger than a second inclination angle of a first virtual line passing a second boundary between the first surface and the second surface and a third boundary between the third surface and the fourth surface with respect to the second surface.

6. The semiconductor device according to claim 5, wherein the detecting element further has a fifth surface extending outward from the fourth surface when viewed in the direction orthogonal to the first surface, and a sixth surface extending from the fifth surface toward the base member.

7. The semiconductor device according to claim 6, wherein, when viewed in a direction along the first surface, the first inclination angle of the virtual tangent to the surface of the recess at the first boundary between the recess and the protective film that covers the second surface with respect to the second surface is larger than a third inclination angle of a second virtual line passing the second boundary between the first surface and the second surface and a fourth boundary between the fifth surface and the sixth surface with respect to the second surface.

8. The semiconductor device according to claim 7, wherein the first inclination angle is larger than or equal to 35 degrees.

9. The semiconductor device according to claim 4, wherein the first inclination angle is larger than or equal to 35 degrees.

10. The semiconductor device according to claim 1, wherein the detecting element further has a fifth surface extending outward from the fourth surface when viewed in the direction orthogonal to the first surface, and a sixth surface extending from the fifth surface toward the base member.

11. The semiconductor device according to claim 10, wherein, when viewed in a direction along the first surface, a first inclination angle of a virtual tangent to a surface of the recess at a first boundary between the recess and the protective film that covers the second surface with respect to the second surface is larger than a second inclination angle of a first virtual line passing a second boundary between the first surface and the second surface and a third boundary between the fifth surface and the sixth surface with respect to the second surface.

12. The semiconductor device according to claim 11, wherein the first inclination angle is larger than or equal to 35 degrees.

13. A method for manufacturing a semiconductor device, the method comprising: forming a first groove in a main surface of a base portion to divide the main surface into a plurality of areas; covering the main surface including the first groove with an insulating protective film; cutting the base portion covered with the protective film along the first groove to divide the base portion into a plurality of detecting elements each having a first surface with a detector; placing each of the detecting elements on a base member; adhering a release film to a die having a cavity with a protrusion; placing the die with respect to the base member while the first surface of the detecting elements is thrust into a portion of the release film on a surface of the protrusion; filling the cavity of the die with a resin material; and detaching the die and the release film from a resin package formed from the resin material after curing, wherein at least a portion of an outer periphery of the surface of the protrusion is located outward from an outer periphery of the first surface when viewed in a direction in which the protrusion of the die and the first surface of the detecting element face each other.

14. The method for manufacturing a semiconductor device according to claim 13, further comprising: after covering the main surface with the protective film and before cutting the base portion, forming a second groove with a smaller width than a bottom surface of the first groove along the bottom surface; and cutting the base portion covered with the protective film along the first groove and the second groove so as to divide the base portion into the plurality of detecting elements.